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SARVAAGAMI



REENIDHI'S ANNUAL TECHNICAL FEST

★ ISO 20121:2012 CERTIFIED

ROBOVEDA'24

EXPLORE THE UNWRITTEN LORE...

SARVAAGAMI

1. GAME AND RULES:

Starting with designing and building a bot that moves through the curves of the track might be easy, but making it run on terrains, hurdles, and being an early bird is a tough task. So are you up for the challenge?

1.1. PROBLEM STATEMENT

1.1.1. Design a manually controlled bot to crack the arena in the least amount of time, avoiding penalties. The arena consists of hurdles that downturn the robot's speed.

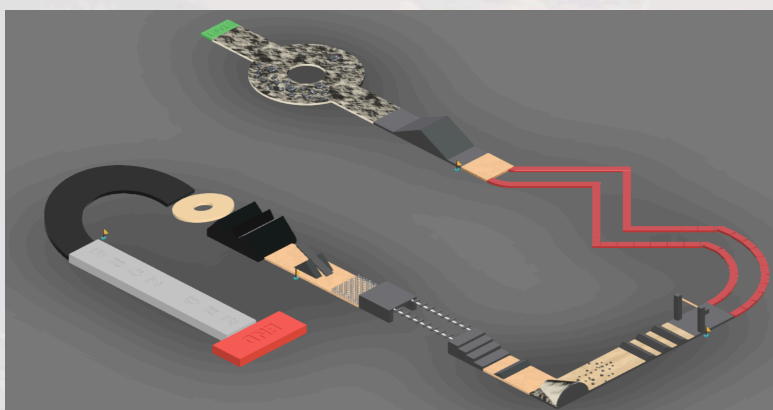
2. GAME FIELD AND OBJECTS:

2.1. ARENA

2.1.1. The arena is placed on the ground, there may be obstacles at certain points in the racecourse.

2.1.2. Minimum track width of the arena is 200mm.

Note: The design is not drawn to scale. The orthographic view of the arena is shown below.



2.2. CHECKPOINTS

- 2.2.1. There are 4 checkpoints in the arena.
- 2.2.2. One at the starting of the zone 2, the other at the end of the gate (zone 2) and another after the nail trap(zone 4), and another at the end of the semi-circular path (zone 5).
- 2.2.3. At any stage of the arena, if the bot is stuck then it will be repositioned at the most recently traversed checkpoint.
- 2.2.4. Checkpoints are indicated by flags in the arena.

3. TEAMS:

- 3.1. There is no team limit.
- 3.2. Only 2 members of a team are allowed to stay around the arena (one for controlling and one for assisting) and a team is allowed to play with one robot only.
- 3.3. Members of different institutions can form a team.
- 3.4. Students who are pursuing any course upto post graduation can participate.

4. ROBOTS:

4.1. ROBOT SPECIFICATIONS

- 4.1.1. Robots can be wired or wireless.
- 4.1.2. Robots should not damage any part of the arena. If the robot damages any part of the arena, it will be disqualified.
- 4.1.3. The robot should not expand or compress during the run. A technical timeout can be taken in case of any part detachment of the robot.
- 4.1.4. A robot is allowed to participate only once in the event. The robot can participate once again with the same team if it satisfies the following conditions:
 - The team has to register once again with a modified old robot/new robot before registrations close and a maximum of 3 registrations will be accepted.
 - Wheels or motors or chassis materials should be changed to consider an old robot as a new one
- 4.1.5. Use of IC engines is not allowed.
- 4.1.6. Readymade PCBs are not allowed.

4.2. ROBOT DIMENSIONS

4.2.1. Dimensions of the robot should not exceed 30cm x 30cm x 25cm (l*b*h). (Tolerance of 5% is allowed in this regard).

4.3. POWER SUPPLY FOR ROBOTS

4.3.1. The potential difference between any two points should not exceed a limit of 12V and the robot should be powered only using batteries. (A tolerance of 5% is allowed).

4.3.2. If the robot suddenly stops in the arena due to any technical problem, you will be allowed to take a technical timeout. After taking one technical timeout, the timer will not be stopped if the robot suddenly stops in the arena due to any technical problem.

4.3.3. The power supply may be ON or OFF the board (i.e., one of the team members can hold the power supply). An external power supply will not be provided.

4.3.4. Changing the battery or battery polarity is not allowed during the race.

4.4. WEIGHT

4.4.1. The maximum weight of the robot should not exceed 5kgs.

5. SAFETY

5.1. All the required accessories should be brought by the participants.

5.2. Team Roboveda is not responsible for any damage to the robot.

5.3. Technical assistance will not be provided by coordinators or technical deputies.

6. THE COMPETITION AND MATCHES:

6.1. MATCHES

6.1.1. A qualification round will be there on the first and second day of the event.

6.1.2. Participants completing the qualification round arena are only shortlisted for the final round.

6.1.3. There will be a final round for participants qualified in the qualification round.



6.2. QUALIFICATION ROUND

6.2.1. This round consists of 5 zones in a curved zigzag manner.

6.2.2. To qualify, the robot has to traverse these zones in a minimum amount of time.

6.2.3. In every zone dimension of properties there might be 5% of tolerance.

6.2.4. ZONE 1

In Zone 1, there will be a sand path in a circular ring, the left side is narrow with small rocks requiring precision, while the right side is wider with large rocks, presenting bigger obstacles but more space to move. After which there is an inclination that is positioned at an angle of 45 degrees at the beginning which is followed by a declination with some obstacles on it.

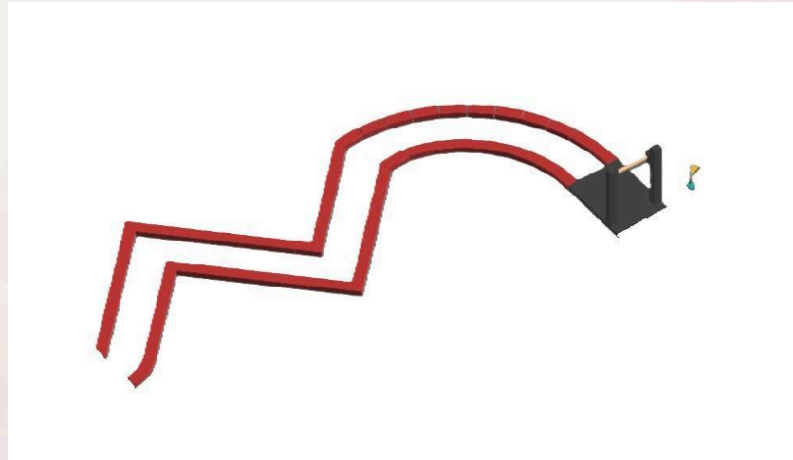


6.2.4.1. Dimensions

The inclination runs 700mm approximately (up to 5% of tolerance) and the same follows with the declination. The height of the zone is approximately 350 mm.

6.2.5. ZONE 2

This zone consists of a Zig-Zag and curved brick path on which the bot has to move forward. Thereafter, a gate is placed which opens manually by button mechanism. Next there is a declination positioned at an angle of 40 degrees at the end of the zone.

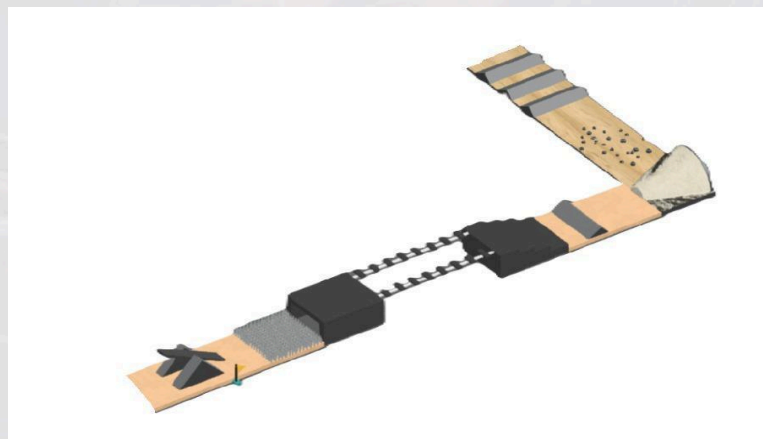


6.2.5.1. Dimensions

The gap between two bricks at either side of the track is approximately 200mm and the height of the declination ramp is 140mm.

6.2.6. ZONE 3

This zone consists of a sandy and rock path containing pebbles of different sizes and speed breakers, followed by a mud mountain which the bot needs to climb and make a turn and cross another speed breaker. It is then succeeded by a series of steps that the bot needs to climb. After climbing the steps, the bot has to move over a hanging bridge followed by a horizontal plank. Henceforth, there is a rectangular plank at a height of 380mm on which the bot has to move and jump onto the nail trap and then pass over the see-saw.

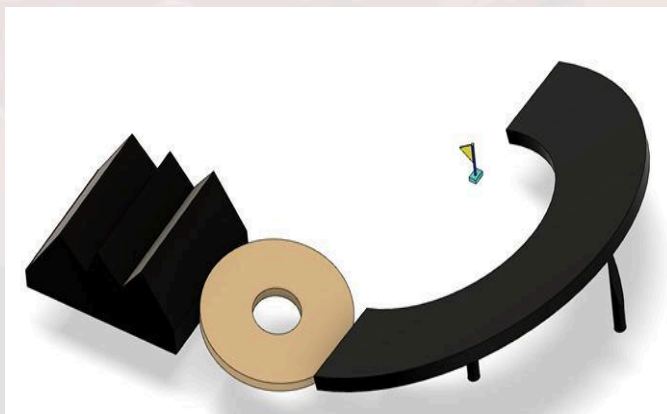


6.2.6.1. Dimensions

The outer and inner radii of the mud mountain are 10.3 cm and 21.5 cm respectively. The length and breadth of the nail trap are 835 mm and 600 mm respectively. The speed breaker has an inclination of 30 degrees and a declination of 45 degrees. The hanging bridge is of length 680mm and width 400mm. The see-saw is of length 90cm and width 35cm.

6.2.7. ZONE 4

In this part of the track, there will be an inclination followed by a crest-trough path and followed by a declination. Next, there is a circular ring followed by a semi-circular path where the bot has to take a curved path to move forward.

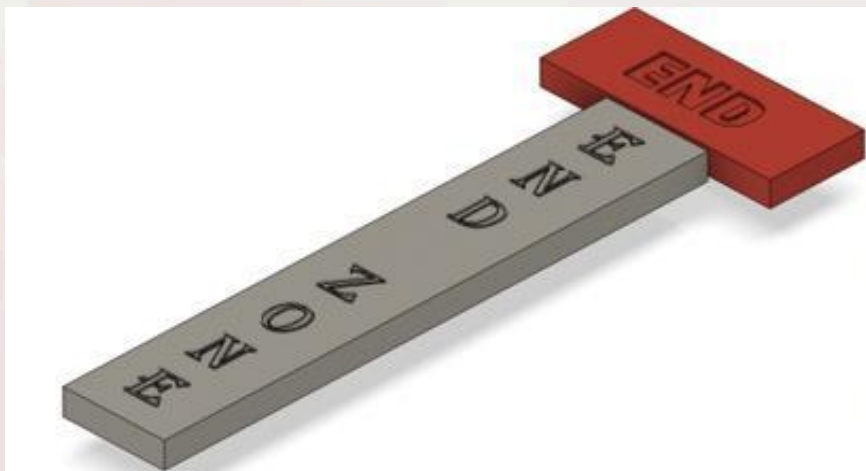


6.2.7.1. Dimensions

Inclination 30 degrees (approximately) the height of the bottom point of the crest-trough path from the ground is 255mm (approximately). The height of the topmost point of the crest from the bottom point is 360mm (approximately) and the top angle is 120 degrees (approximately), declination 44 degrees (approximately). The inner radius and outer radius of the circular ring are 194mm and 388mm (approximately) respectively. The circular ring is placed at an angle such that the height is 214 mm. The parabolic path runs for a length of 1080mm.

6.2.8. ENDZONE

This is going to be revealed on the day of the event.



7. COMPETITION RULES:

7.1. TECHNICAL TIMEOUTS

- 7.1.1. A technical timeout of 1 minute can be taken by a team.
- 7.1.2. After the finishing of the technical timeout, the timer will start again and will not be stopped under any circumstances.
- 7.1.3. No technical assistance will be provided by the event coordinators or heads during the event.

7.2. SCORING & PENALTIES

- 7.2.1. The score will be calculated based on the time taken to complete the track.
- 7.2.2. Based on the violations and penalties the changes in the score are made.
- 7.2.3. The maximum time for each participant is 12 minutes for track completion.
- 7.2.4. The team which completes the qualification round in the least amount of time with the least no. of penalties is eligible for the next round.
- 7.2.5. Scoring schema (including penalties) will be disclosed on event day by heads and coordinators.

7.3. OTHERS

- 7.3.1. Human interference can be only done during skip, technical timeout, and checkpoints.
- 7.3.2. If the robot engages in a place without any further movement, then it would be repositioned at the nearest checkpoint traversed.
- 7.3.3. Participants with wired robots are strictly advised to get wires of length 3m or more.
- 7.3.4 In the case of wired robots, the wired should be slacked throughout the game.

8. DISQUALIFICATION:

- 8.1. If a participant does not comply with the rules of the event, the robot will be disqualified from the event.
- 8.2. Robots should not damage any part of the arena. If the robot damages any part of the arena, it will be disqualified.
- 8.3. Human interference (e.g. touching the robot) during the game will be disqualified.
- 8.4. Misbehavior of any kind will not be tolerated and the team will be subjected to disqualification from Roboveda.
- 8.5. At any point of time during the match voltage should not be varied and should be constant, variation in voltage leads to disqualification.
- 8.6. A robot with the base of a toy car and its gearbox as a machine part will be disqualified.
- 8.7. Also, Lego kits are strictly prohibited and will lead to disqualification.
- 8.8. Robot unable to complete the track traversal within time (12 minutes) is considered disqualified.
- 8.9. Using any of the checkpoints more than 3 times (total) is considered disqualified.

9. COMPETITORS AND BEHAVIORAL RULES:

- 9.1. All participants will be given a Certificate of Participation.
- 9.2. Winners and runners will be given a Certificate of Merit.
- 9.3. All participants must bring both a college ID card and an event ID card.
- 9.4. Participants are allowed to participate once again if they satisfy rule 4.1.4.
- 9.5. Participants are not allowed to use their own Stop watches.
- 9.6. Lending or making the robot participate in the same event with a different team will not be entertained.

9.7. The dimensions, weights, etc., of the field, facilities and equipment described in this document have a margin of error of plus or minus 5% unless otherwise stated.

9.8. Team Roboveda is not responsible for any kind of damage to your robot.

9.9. We request the participants not to assume anything that is not mentioned in the document without contacting the Coordinators or Technical team.

9.10. It's our sincere request to all the participants to go through this rule book thoroughly and not a single point mentioned in the rule book will be compromised under any circumstances. We also request you have a copy of this rule book handy during the event.

9.11. In case of any discrepancies, the decision of co-ordinators and heads shall be final and no further argument will be entertained.

9.12. Make sure you visit our official website on a daily basis so that you will get to know if there are any changes made in the event.

9.13. Any kind of misbehavior will not be tolerated and will lead to disqualification of the team.

9.14. If you choose our hospitality, the payment must be made on the day you check-in.

9.15. In unforeseen circumstances, Team Roboveda abides by the decision of the Institution regarding the conductance of the event. Only registered participants will be informed about the change in event dates or cancellations. In that case, the registration fee will be refunded.

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